

ASSIGNMENT 2 — Synchronization & Explicit Waits

1. Experiment Title

Synchronization Using Selenium Explicit Waits

2. Problem Statement

Automate a web page containing dynamically loaded content. The script must wait for the required element to become visible before extracting its text. The use of `time.sleep()` is prohibited.

3. Objective

- To understand synchronization in Selenium.
- To implement `WebDriverWait`.
- To use `expected_conditions`.
- To wait dynamically for an element.
- To extract text only after the element becomes visible.

4. Tools, Software and Concepts Used

Tools & Software:

- Python
- Selenium WebDriver
- Google Chrome
- VS Code / PyCharm

Concepts:

- Synchronization
- Explicit Wait
- `WebDriverWait`
- `expected_conditions`
- `visibility_of_element_located`

- Dynamic Web Elements
- Text Extraction

5. Source Code / Implementation Steps

```
from selenium import webdriver

from selenium.webdriver.common.by import By

from selenium.webdriver.support.ui import WebDriverWait

from selenium.webdriver.support import expected_conditions as EC

def test_explicit_wait():

    driver = webdriver.Chrome()

    driver.maximize_window()

    driver.get(

        "https://the-internet.herokuapp.com/dynamic_loading/1"

    )

    start_button = driver.find_element(By.XPATH, "//button")

    start_button.click()

    wait = WebDriverWait(driver, 10)

    text_element = wait.until(

        EC.visibility_of_element_located(

            (By.XPATH, "//div[@id='finish']/h4")

        )

    )

    text = text_element.text

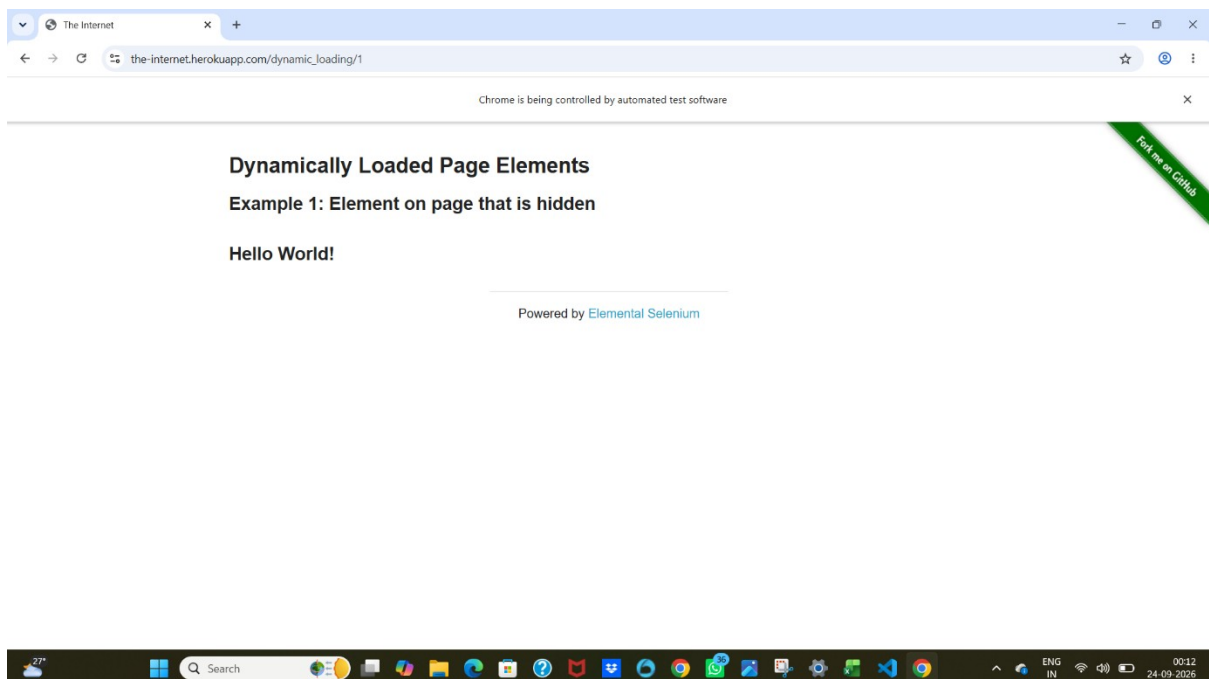
    print("Extracted Text:", text)

    assert text == "Hello World!"

    wait.until(
```

```
EC.visibility_of_element_located(  
    (By.XPATH, "//*[@id='finish']/h4")  
)  
)  
  
driver.quit()
```

6. Output



7. Result

The dynamic content was successfully detected and extracted using WebDriverWait and expected_conditions, without using time.sleep().

8. Observation

- Explicit waits synchronize automation with dynamically changing web elements.
- The script waits only until the required condition is satisfied.
- visibility_of_element_located verifies that the element is visible before extracting its text.

9. Conclusion

This experiment demonstrated effective Selenium synchronization using **explicit waits**, providing a more reliable approach for handling dynamically loaded web content.